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Causal ML: Python package for causal inference machine learning

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ABSTRACT

“Causality” is a complex concept that is based on roots in almost all subject areas and aims to answer the “why” question. Causal inference is one of the important branches of causal analysis, which assumes the existence of relationships between variables and attempts to examine and quantify the actual relationships in the available data. Machine learning (ML) and causal inference are two techniques that emerged and developed separately. However, there is now an intersection between these two fields. Causal ML is a Python package that provides a set of uplift modeling and causal inference methods using machine learning algorithms based on recent research. It gives the user a standard interface that lets them estimate conditional average treatment effects (CATE) or individual treatment effects (ITE) based on experimental observational data.

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Code metadata

Current code version	1.0
Permanent link to code/repository used for this code version	https://github.com/ElsevierSoftwareX/SOFTX-D-22-00286
Permanent link to reproducible capsule	
Legal code license	Apache License, Version 2.0
Code versioning system used	Git
Software code languages, tools and services used	python
Compilation requirements, operating environments and dependencies	Python 3.6, 3.7, 3.8 and 3.9
If available, link to developer documentation/manual	https://CausalML.readthedocs.io/en/latest/index.html
Support email for questions	

1. Motivation and significance

Nowadays, the study of causality covers a wide range of disciplines, such as economics, law, medicine, environmental science, computer science, and philosophy [1–3]. In almost all fields of research, there are numerous “why” questions, which are the path to the future of science, and causal analysis aims to explain the why. In the field of causal analysis, two main tasks can be distinguished: causal discovery and causal inference. Causal discovery is responsible for analyzing and creating models that account for the relationships inherent in the data. While the

purpose of causal inference is to examine the possible effects of intervening in a particular system [4].

The Potential Outcome Model [5] and the Structural Causal Model (SCM) [6] are two popular models in the field of causal inference. Both of the two frameworks are essentially the same [7], in that they infer causality in observed data but use different ideological principles. The Potential Outcome Model defines a causal effect as the difference between potential outcomes for the same subject [5], and links the intervention to the research object. The intervention is the cause, and the result of the intervention is the effect [8]. Compared with the Potential Outcome Model, the SCM is more accurate in determining confounders [9] and is becoming more and more popular. The SCM uses graph theory, a mathematical tool, to formally express the causal assumptions behind the data. In contrast to the potential outcome model, the structural causal model approach requires knowledge of the

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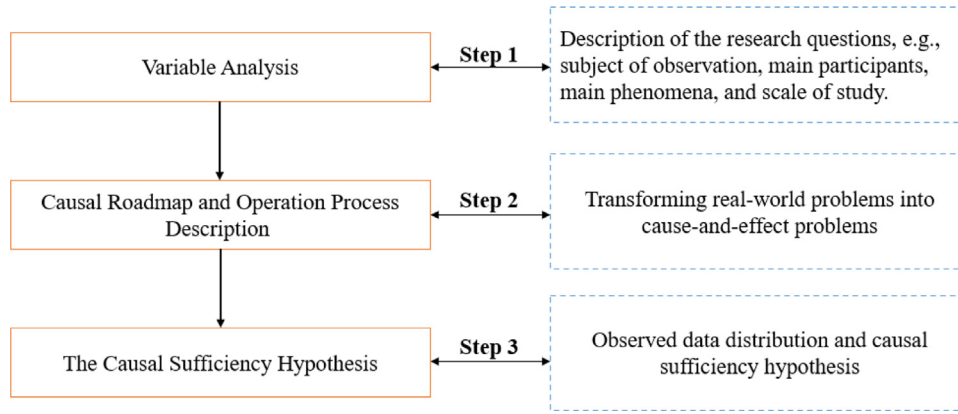


Fig. 1. Causal Sufficiency Analysis.

existing causal model but does not make strong assumptions about the model form.

Although correlations can be viewed as the result of underlying causal mechanisms [10], however, as expressed by a well-known saying in causal analysis, correlations do not imply causality [11]. Additional assumptions about the underlying causal structure are often necessary to guide inference based on observed data before causal inference can be made.

In [12], this process of establishing causality is divided into 3 stages: analysis of the variables involved in the research question; analysis of the causal map and causal mechanisms; and causal adequacy assumptions. In [13], a causal analysis approach based on causal maps was further elaborated with reference to the studies of others [14–17]. [13] argue that in causal analysis, it is necessary to first identify the research questions, including the object of observation, the main participants, the main phenomena, and the size of the examination. Then a causal model is specified and the research question is transformed into a causal problem. Finally, the variables' causal links must be investigated, recognized, and postulated. As shown in Fig. 1, the analysis process of [13] is consistent with [12]. The absence of an adequate causal analysis prior to causal inference will lead to fallacies in the results, which [18] describes as "causal model misspecification". [13], on the other hand, argues that such errors are due to incorrect causal models, and therefore such errors should be called "cause model neglect": i.e., a failure to use causal knowledge to carefully specify the target statistical parameters before performing estimation.

Most of the time, the work of causal inference starts from the premise that the causal variables are given [19]. The task of causal inference is founded on the concept of causal sufficiency, which is used to assess the causal influence of a specific variable (treatment) on an outcome of interest and thus causally explain the complete causal model [11,13]. There are many ways to estimate causal effects, e.g., with matching, G-computation, inverse weighting, or augmented inverse weighting. The two fields of causal analysis and machine learning arose and developed separately. However, in recent years, the two fields have intersected. It has been found that statistical-based AI is not intelligent because statistical models cannot generalize, whereas causal models allow for modeling distributional changes through the concept of intervention [7,20]. Similarly, causal analysis can also benefit from machine learning. While traditional causal analysis is based on statistical models, current machine learning is also based on statistical models. Supervised learning gives us a way to do this kind of exploration in a way that is strong, logical, and fully planned out. This can make up for the problems with traditional causal inference [13].

Causal ML is a Python package that offers a suite of mathematical modeling and causal inference tools based on cutting-edge

research [21]. Causal ML is mainly based on Structural Causal Model to observe causal effects from the data. However, it also uses some Potential Outcome Model approaches such as Matching [22]. It provides a common interface through which users can estimate conditional average treatment effects (CATE) or individual treatment effects (ITE) using experimental or observational data.

2. Software description

2.1. Software architecture

2.1.1. Algorithms

There is a lot of exploratory research on the use of ML for causal inference [23,24], and there are some important practical applications in these studies. Causal ML aims to build a one-stop store of machine learning for causal inference. This is an approach that can play an important role in business, science, and other fields. It can democratize ascending modeling approaches that currently exist only in academic papers or different statistical packages. The current version of Causal ML implements 15 state-of-the-art uplift modeling algorithms (See Fig. 2). With its good generality and ease of use, Causal ML is an excellent choice for causal inference.

For reasons of manuscript length and structure, we present only some of the algorithms in Fig. 2. The Meta-Learner Algorithms build on the base algorithm to estimate the ATE [23, 25]. Theoretically, meta-algorithms can use any base learner, such as random forests (RFs), Bayesian additive regression trees (BARTs), or neural networks. Causal ML supports five main meta-algorithms: the S-learner, the T-learner, the X-learner, the R-learner, and the Doubly Robust (DR) learner. Each meta-algorithm has a different way of estimating the average output and defining CATE. S-Learner and T-Learner are used to show the differences between them.

The S-learner uses a single machine learning model to estimate treatment effects, as follows:

Step 1 Using a machine learning model to estimate average outcomes $\mu(x)$ using covariates X and a treatment effect indicator variable W :

$$\mu(x) = E[Y|X=x, W=w]. \quad (1)$$

Step 2 The estimated value of CATE is defined as:

$$\hat{\tau}(x) = \hat{\mu}(x, W=1) - \hat{\mu}(x, W=0). \quad (2)$$

The T-learner is simple and easy to understand, and it trains two base learners (the name "T" comes from two base models) for

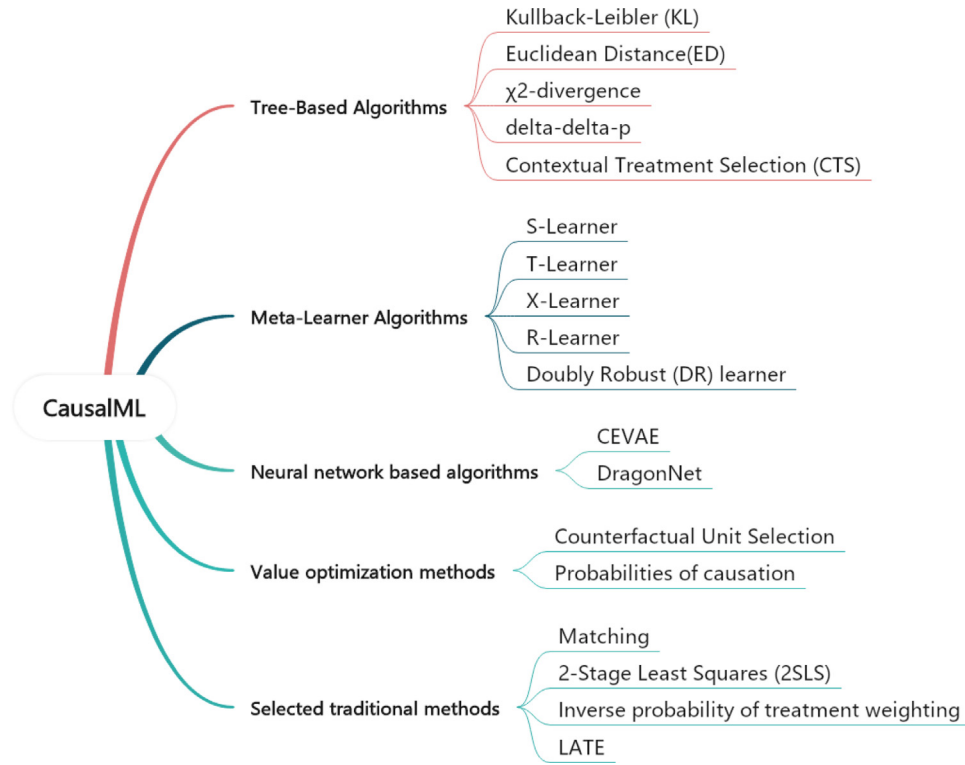


Fig. 2. Causal ML Algorithm Diagram.

the control and treatment groups. The estimation process consists of two steps.

Step 1 Estimate the average output of the models $\mu_0(x)$ and $\mu_1(x)$:

$$\mu_0(x) = E[Y(0)|T=0], \quad (3)$$

$$\mu_1(x) = E[Y(1)|T=1]. \quad (4)$$

Step 2 The estimate of the average treatment effect (ATE) is defined as

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x). \quad (5)$$

The random forest algorithm, proposed by [26], is a widely used statistical learning algorithm. Statisticians usually study random forests as a practical method for nonparametric conditional mean estimation [27]. Causal ML supports boosted tree-based estimation of CATE. The boosted tree method is a tree of methods that use a tree-based algorithm and split based on the difference in boosts. [28] proposed three different methods to quantify the divergence gain due to splitting [29].

$$D_{\text{gain}} = D_{\text{after split}}(P^T, P^C) - D_{\text{before split}}(P^T, P^C), \quad (6)$$

where P^T and P^C stand for the probability distributions of the desired outcome in the treatment and control groups, respectively, and D denotes divergence. Causal ML implements several different methods for quantifying divergence, such as Kullback–Leibler (KL), Euclidean Distance (ED), and Chi, etc.

The Kullback–Leibler (KL) divergence is given by the following equation:

$$L(P:Q) = \sum_{k=\text{left, right}} p_k \log \frac{p_k}{q_k}, \quad (7)$$

where p is the sample mean of the treatment group and q is the sample mean of the control group, and, indicates the leaf in which p and q are computed (Liu & Shum, 2003).

The Euclidean Distance (ED) is given by the following equation:

$$D(P:Q) = \sum_{k=\text{left, right}} (p_k - q_k)^2, \quad (8)$$

where the symbols are the same as in the above equation.

The general idea of matching methods is to find treatment and non-treatment units that are as similar as possible in terms of relevant characteristics. Thus, matching methods can be seen as part of a family of causal inference methods that attempt to mimic randomized controlled trials. Many scholars have explored causal inference pathways based on matching methods. [30] addressed the problem of matching methods by redefining the matching problem as a subset selection problem. [31] proposed a new approach to matching in observational studies. The essential ideas of this novel method of matching are to balance the treatment groups with respect to a target population and to formulate the matching problem as a linear-sized mixed integer.

Although there are several techniques to match treated and untreated units, the propensity score is the most widely used approach:

$$e_i(X_i) = P(W_i = 1|X_i), \quad (9)$$

Following that, treated and untreated units are matched in terms of $e(X)$ using a distance criterion, such as $k:1$ nearest neighbors. Since the treatment group and the control population are often matched, this method estimates the average treatment effect on the treated (ATT).

$$E[Y(1)|W=1] - E[Y(0)|W=1] \quad (10)$$

The advantages and disadvantages of different matching methods are discussed in [32].

The evolution of causal ML has paralleled the continued maturation of causal inference theory and machine learning causal inference. Although Causal ML contains many mature causal

inference algorithms, such as tree-based algorithms and meta-learning-based algorithms, it is not perfect, and many of the latest algorithms have not been implemented. However, it is not perfect, and many of the latest algorithms have not yet been implemented. The reasons for this are related to both Causal ML itself, such as the enthusiasm of the contributors and the scope of their knowledge, as well as the maturity and industrialization of the algorithms. Causal ML does not usually go ahead and use an algorithm first. Instead, it waits until other platforms have shown that the algorithm is ready and reliable.

2.1.2. Package

Table 1 shows the main package structure of Causal ML, the details of which are available on the official website of Causal ML. In Causal ML, the initial release (V0.2.0, 2019-08-12) mainly supports causal inference based on uplift trees and meta-learning algorithms. These algorithms were placed under the *causalml.inference.tree* and *causalml.inference.meta* packages, and their parent package was *causalml.inference*. In versions V0.7.0 (2020-02-28) and V0.8.0 (2020-07-17), Causal ML updated the *nn* subpackage and *iv* subpackage of *causalml.inference* to support neural network-based and 2SLS-based inference algorithms, respectively. It is easy to see that *causalml.inference* was designed at the beginning mainly for organizing and archiving algorithms belonging to Structural Causal Model (SCM) [6], while algorithms based on Potential Outcome Model [5] were not placed under the *causalml.inference* package, for example, matching algorithms were placed under the *causalml.match* package.

The *causalml.features*, *causalml.dataset*, and *causalml.metrics* packages were also present at the beginning of the release of Causal ML, and these packages provide supporting work for data and feature processing and evaluation necessary for causal inference to be performed. *causalml.feature_selection* is another supporting toolkit updated in Version 7.0 (2020-02-28) for interpreting the results of causal inference.

Since causal inference machine learning is still a rapidly evolving branch of technology and Causal ML is a young scientific tool, there are some implausibilities in its structural organization. For example, matching-based, propensity score, and 2SLS are some traditional causal inference tools, and only the 2SLS method is placed under the *causalml.inference* package. If this is done to tell the difference between Structural Causal Model and Potential Outcome Model, algorithms that are not part of the Structural Causal Model are not put in the same package. Another example is NearestNeighborMatch which is a Propensity Score Matching type of method placed under the *Causal ML.match* package. ElasticNetPropensityModel is a Propensity Score Estimation method that is placed under the *Causal ML.propensity* package. This appears to be a reasonable but not very well justified way of laying out the facts. The difficulty in explaining these organizational structures is certainly related to the rapid growth of the subject area and the collaboration of multiple developers. But it also shows that the way the software is set up has not been thought through well enough. But this cannot be fixed overnight, and we think these problems will be solved well as knowledge of causal inference machine learning grows and versions are updated.

2.2. Software functionalities

Essentially, Causal ML estimates the effect of intervention T on the causal relationship $X \rightarrow Y$ without making strong assumptions about the form of the model. Causal ML implements multiple treatment effect evaluations such as average treatment effect estimates (ATE), conditional average treatment effect (CATE) estimates, and individual treatment effects (ITE). ATE and CATE are based on total treatment effect (TTE) estimates and conditional

total treatment effect estimation (CTTE). TTE can be defined as the comparison of a binary treatment T to Y interventions:

$$TTE = E[Y|T = 1] - E[Y|T = 0]. \quad (11)$$

CTTE can be defined as the comparison of interventions of a binary treatment T on Y , with $X = x$ is satisfied:

$$CTTE = E[Y|T = 1, X = x] - E[Y|T = 0, X = x]. \quad (12)$$

In the process of supervised learning, prediction models can evaluate the importance of feature variables based on their contribution to the prediction performance. Based on this property, Causal ML provides causality analysis that can be explained. Causal ML can evaluate the importance of $x_0, x_1, \dots, x_m$ in the process of quantifying the effect of X on Y ($X \rightarrow Y$) by considering a set of feature combinations $X = [x_0, x_1, \dots, x_m]$ consisting of multiple feature variables.

Causal ML also provides methods to interpret the trained treatment effect models, such as Meta-Learner Feature Importances Visualization, Uplift Tree Visualization, and Uplift Tree Feature Importances Visualization.

Because the true value is not known except for the experimental data, estimation of the treatment impact cannot be validated in the same manner as ordinary ML predictions. Here, we concentrate on internal validation approaches based on the premise of unconfoundedness of potential outcomes and treatment status based on the feature set at our disposal. Causal ML currently supports validation with multiple estimates, validation with synthetic data sets, validation with uplift curve (AUUC), and validation with sensitivity analysis validation methods.

3. Illustrative examples

We use an example of evaluating the impact of holidays on investor sentiment to show how causal effects can be assessed using Causal ML. Both traditional statistics-based causal estimation and machine learning-based causal estimation are based on determined causal sufficiency (Mooney et al. 2021). In contrast, real-world problem analysis and causal adequacy assumptions rely almost exclusively on human knowledge and expertise. Neglecting causal adequacy analysis will lead to causal model errors, thus rendering causal explanations meaningless.

In our case, we use the general context of the Wuhan city closure at the beginning of the COVID-19 outbreak as an intervention. With reference to [33], we analyzed the impact of Wuhan City's closure on stock prices and investor sentiment. This paper constructs an investor confidence index (ICI) as a proxy for investor sentiment based on investor messages on social media with reference to the method of Antweiler & Frank (2004). And, as [34] suggested, the agreement index was not created.

At the beginning of the study, we must make the necessary elaborations and causal assumptions about the real-world problem.

3.1. Relevant variables

We consider 2 variables P^d , ICI^d and a treatment T to evaluate the effect of city closure on the causal relationship between stock price and investor sentiment.

(1) P^d is the stock price fundamental at date d . P^d consists of the opening price p_o^d , the closing price p_c^d , the highest price p_H^d and the lowest price p_L^d , $P^d = \{p_o^d, p_c^d, p_H^d, p_L^d\}$.

(2) The investor confidence index ICI^d for date d is obtained by calculating the number of positive messages $N_{Positive}$ and the number of negative messages $N_{Negative}$ among the messages published by investors on social media on date d .

(3) Treatment T : We see the closure of the city as an intervention.

Table 1
The main packages of Causal ML.

Package	Parent package	Package description	Method examples
Causal ML.inference.tree	Causal ML.inference	Package of tree algorithms.	UpliftTreeClassifier UpliftRandomForestClassifier
Causal ML.inference.meta	Causal ML.inference	Package of meta-learning algorithms.	LRSRegressor XGBRegressor, MLPTRRegressor
Causal ML.inference.iv	Causal ML.inference	Package based on 2SLS method.	IVRegressor
Causal ML.inference.nn	Causal ML.inference	Package of neural network algorithms.	CEVAE
Causal ML.optimize	Causal ML	A package of optimization methods using counterfactual unit selection and a counterfactual value estimator.	CounterfactualValueEstimator PolicyLearner get_actual_value
Causal ML.match	Causal ML	Package based on matching methods	NearestNeighborMatch MatchOptimizer
Causal ML.propensity	Causal ML	Package based propensity score model.	ElasticNetPropensityModel
Causal ML.feature_selection	Causal ML	A class for feature importance methods.	FilterSelect
Causal ML.dataset	Causal ML	Package providing simulated data.	make_uplift_classification synthetic_data
Causal ML.metrics	Causal ML	Package for causal inference effect validation or to assist in causal inference effect validation.	Sensitivity SensitivitySubsetData
Causal ML.features	Causal ML	A package that handles label vector	LabelEncoder OneHotEncoder

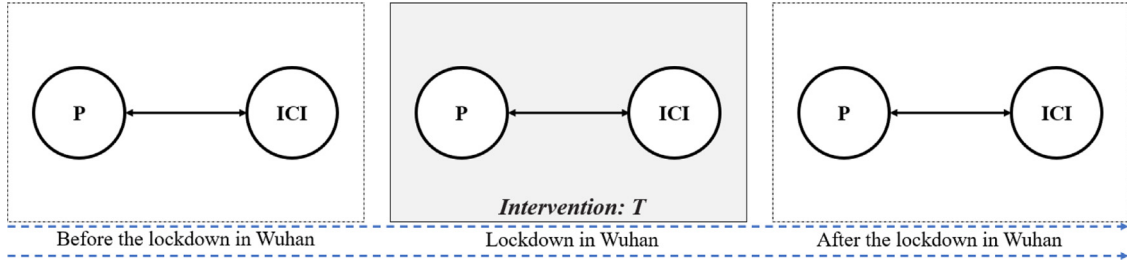


Fig. 3. Assumed causal graph.

3.2. Causal maps

We assume a causal graph as shown in Fig. 3, where stock market fundamentals P and investor sentiment ICI are causally related to each other.

(1) Fundamentals of stock prices at time t_0 (P^{t_0}) influence investor sentiment at time t_1 (ICI^{t_1}),

$$P^{t_0} \rightarrow ICI^{t_1}.$$

(2) Investor sentiment ICI^{t_1} at t_1 influences stock market performance P^{t_2} at t_2 , $ICI^{t_1} \rightarrow P^{t_2}$. And t_0 , t_1 and t_2 can be very close to each other. The interaction between the stock price P and the investor sentiment ICI is high-frequency [35–37].

(3) An extraneous change in the environment interferes with the causal mechanism of $P \leftrightarrow ICI$. In this case, we assess the impact of the external intervention of the closed city on the causal mechanism of stock price and investor sentiment.

3.3. Causal sufficiency

In this example, we assume that the fundamentals of the stock market (P) and how investors feel about the market (ICI) are related in a causal way. Our hypothesis is based on studies by others. [38] argue that investor sentiment is an investment's estimate of future earnings and cash flows. Current stock market performance affects investors' estimates, so $P \rightarrow ICI$. And investor sentiment likewise affects stock market fundamentals [38–40], so $ICI \rightarrow P$. We assume that there are no unobserved confounding factors, i.e., that $ICI \leftrightarrow P$ is causally sufficient. Although this is a strong assumption, it is necessary for reasoning about causal effects [12].

We consider the environmental context of the city's closure as an intervention. The effects of the city closure on stock price (P) and investor sentiment (ICI) are estimated by assessing the causal effects of stock price (P) and investor sentiment (ICI) before, during, and after the city closure.

We use T-learner, a meta-learning algorithm, to perform causal effect estimation. We estimate the effect of city closure on the causal effect between stock price fundamentals and investment sentiment ($P \leftrightarrow ICI$) using LGBMRegressor, XGBRegressor, and RandomForestRegressor as the underlying learners, respectively. As shown in Table 2, the average disposition effect of city closure on $P \leftrightarrow ICI$ is 0.7516, which implies a higher synergy between stock price and investment sentiment during city closure. Investors are more sensitive to changes in stock prices during the city's closure period.

4. Impact

Causal ML applies the latest advances in the field of causal inference machine learning from academic papers to the real world. It facilitates the practical application of machine learning in the field of causal inference by building a one-stop shop for machine learning causal inference. Causal ML was released in its first public version in August 2019 and has been updated 16 times.

Causal ML received a lot of attention from the academic community upon its publication. [4] listed Causal ML as one of the four currently available tools for causal inference. Another causal inference tool recommended by Yao et al. DoWhy, supports integration with EconML and Causal ML packages [41]. The Causal

Table 2
The causal effect of lockdown in Wuhan.

	LGBMRegressor	XGBRegressor	RandomForestRegressor	Mean
E	0.5784	0.8658	0.8106	0.7516

Note: The range of investor sentiment for the statistical sample is $[-2.145, 2.124]$.

ML was used in the study by [42] to construct uplift trees. An overview by [43] summarizes the latest causal methods, which includes an introduction to Causal ML. [44] used Causal ML to study the causal effect of dynamic quarantine policies on COVID-19.

In summary, Causal ML is a mature Python package for causal inference that has been adopted by numerous studies and is of great academic value.

5. Conclusions

In this paper, we introduce Causal ML, an open-source Python package that can be used for causal inference. It widely incorporates the latest research results in causal inference machine learning and applies these methods, which exist in the academic literature, to practice. It provides a unified interface for causal inference machine learning and a one-stop integration solution, which is very important for academic research.

We provide a simple case that illustrates the process of causal inference using Causal ML. It is important to emphasize that failure to perform adequate causal analysis prior to causal inference may lead to “causal model misspecification” [18].

The latest version of Causal ML is 0.13.0 (updated in September 2022) and is still a causal inference tool in its infancy. As such, it still has many imperfections. Causal ML has not evolved in isolation, and its development is closely linked to that of other open-source machine learning products, such as scikit-learn (<https://scikit-learn.org/stable/>). Since Causal ML tends to use some well-established algorithms that have been validated by other software, some advanced algorithms such as GANITE (a causal inference method based on generative adversarial networks for learning the uncertainty of counterfactual distributions [45]) remain unimplemented. For a rapidly evolving scientific tool, Causal ML certainly has some problems, but many of them cannot be solved overnight and depend on the contributions of a larger number of scholars and the joint progress of academia and industry. However, Causal ML is maturing in the next iteration as more and more positive academic results are absorbed by Causal ML. We believe that Causal ML will become one of the best uses of machine learning for causal inference.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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